

Project Grand Path

An Inclusive, Interdisciplinary, Collaborative, Project Based Learning Model

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ABSTRACT

Project Grand Path (PGP) is leading the way to re-envision higher education in the 21st century by ‘mobilizing learners to be the architects of their education’. Located in the University’s FuturEDlab, a resource for the campus at large, this interdisciplinary, collaborative, student centered. innovation hub is infused with a series of high impact learning experiences, accessible through a curated studio-based internship program. Utilizing design thinking, cutting-edge computer graphics and a mentorship program, PGP fosters an environment of inclusion and equity providing the opportunity for the next generation to dream up a new paradigm for the college experience.

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1 INTRODUCTION

1.1 Overview

Project Grand Path (PGP) was designed with the intent to cultivate the region’s first learner inspired community for educational equity. Through contributions from the College of Engineering and Computing, the Department of Visual and Media Arts and the Information Technology Innovation and Research Ecosystem, internships were designed for college students with the intent to build teams from various disciplines who could work together to bring to fruition provocations from high school students across the United States who are part of the REP4 (Rapid Education Prototyping) program. A multi-disciplinary, high tech studio environment has been created on campus, fostering a bridge between the classroom and industry. This initiative not only bridges connections between students but also between multiple colleges, departments and faculty and staff to foster a culture which paves the way for the eventual development of a larger innovation ecosystem.

1.2 Objective

The objective is to empower the student and the student voice as a method for shaping the evolving higher education experience.

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Utilizing a high impact, project-based learning environment with access to the latest developments in computer graphics, PGP works as a bridge between the traditional classroom, various departments, and the post graduate experience. It is designed to equip students with the tools and soft skills to navigate a landscape where the nature of ‘work’ continues to evolve in industries which have yet to be defined. Mentorship from faculty, staff and industry professionals provides insight into a range of perspectives. The students transform into experienced designers with a foundational framework, inspiring them to implement a fresh vision for the future of education within the world of emerging technology.

2 PROCESS

2.1 Timeline

Over the course of two 15-week semesters and one 12-week summer semester 10-20 students are oriented to learner driven provocations which they take through various weekly sprints and production pipelines. Utilizing a framework of fluid design thinking processes, the implementation of empathy works as a conceptual framework, as problems are defined, and approaches are ideated upon. Teams of 3-5 students are formed and oriented to a range of end user experiences and techniques to consider before entering the prototyping phase. The unique makeup of the student cohort, in conjunction with diverse mentoring, allows for ideas to be infused with techniques and become viable portals for a range of solutions.

2.1.1 Winter 2024 Student Cohort. This experiential approach to learning provides the opportunity to empower the students with the ability to dive in and explore with childlike wonder the tools which have previously been available only to a few. Exposing interdisciplinary teams to a culture of experimentation with various forms of emerging technology becomes transformative. This sensibility emphasizes a cohesive perspective of creating, evaluating, analyzing, applying, and understanding through a high impact experience with evolving interfaces. The result is a framework of openness, which feels accessible to a range of learners and diverse perspectives. An anonymous feedback system is made available at all stages, designed to empower the student voice on any aspect of the process with an inclusion and equity mindset.

2.1.2 Money Mindset. Figure 2 is a screenshot of the “Money Mindset” components on the Project Grand Path portfolio webpage.

2.2 Innovation

At the end of a semester cycle, working prototypes have been tested with user feedback to consider reiterating on, documenting and/or bringing to market. In the first year, the students had two provocations, one of which was to create an interface to make the topic

Summary	Collaborative, interdisciplinary, project-based learning with a focus on design thinking and CGI.
Learning Outcomes	Interdisciplinary soft skills, design thinking, prototyping, pre-viz, app development.
Classification(s)	The curricular topic addressed by this experience is internship and/or senior projects.
Audience	Undergraduate students at all levels and majors from freshmen to seniors
Dependencies	Self-selecting students who desire hands-on experience that will bridge skills with applied experience with critical thinking integrated.
Prerequisites	Desire to contribute to a student driven project. e.g., project dev runs in 3 stages.
Strengths	Unique applied experience developing tech to influence the future of education.
Weaknesses	Getting comfortable with the unknown and working with iterations of prototypes.
Variants	Students can choose 1 semester or stay on for multiple iterations of the design process.
Assessment	Engagement, openness to experience, continuation through cycles, prototype functionality and engagement.



Figure 1: Group photo of Winter 2024 PGP Faculty, Staff and Interns From left to right: (back row) Julie Goldstein (Faculty), Jovan Hodo, Alexa Shiesler, James Johnson III, Bobby Thruong, Calab Bahlbi, Amanda Clouse (Staff) (front row) Alex Cersosimo, Emily Mosier, Saud Alomar, Abby Haywood



Figure 2: The “Money Mindset” is a quiz developed by the first PGP cohort. (Design: Emily Mosier, Illustration: April Nunn, Dev Ops: Andrea Gonzalez, Content Expert: James Johnson IV)

of financial literacy more accessible to the incoming college freshman. The students insisted that before they could make something comprehensive, they needed to understand their own relationship to money. A standalone app was created to access one’s ‘Money Mindset’ via animated avatars, representing various personality profiles, which are revealed in a tarot spread with the data quantified for each user. The success of this fresh prototype has led to additional investment and the opportunity to develop a more extensive framework for learning and peer mentoring.

PGP is also working to develop an empathetic LLM, designed to encourage future learners to access their ‘hero’s journey’ as a paradigm for understanding their educational journey. This provocation began with technical explorations of Open AI’s LLM in conjunction with UI interface and animated avatar options. Tests are being conducted with NVIDIA Omniverse’s open USD solutions and a volumetric digital twinning system alongside traditional 3d and 2d animation methodologies. When in the ideation phase, it is eventually up to the students to choose the method/s to achieve their prototype/s and incorporate the aesthetic and technical considerations into their design document/s which are reviewed by a panel of mentors and advisors.

3 DOCUMENTATION

Documentation on the inception of the PGP structure and student prototypes is available for viewing at <http://www.projectgrandpath.com> along with a range of supplementary materials.

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